

EEI6373 – Performance Modelling

Mini Project Report

Supermarket Cashier Checkout System



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1.Introduction

A supermarket checkout system is a service-based queuing system where customers wait for available cashiers after completing their shopping. Its performance depends mainly on customer arrival rates, the number of available cashiers, and cashier service times. During busy periods, a high number of customer arrivals can result in longer queues and increased waiting times.

This project evaluates the Supermarket Cashier Checkout System using a Discrete Event Simulation model. The simulation represents customer arrivals, cashier queues, checkout service, and customer departures over a fixed operating period.

The main objectives are to analyze customer waiting time, system throughput, queue length, and cashier utilization. The simulation is also used to evaluate how different customer volumes affect the performance of the checkout system.

1. System Description and Performance Goals

The supermarket cashier checkout system is a service setup where customers are attended to by cashiers upon completing their shopping. Customer flow and checkout processes have been simulated to understand how fluctuations in customer demand, the number of items, and cashier availability impact system efficiency.

2.1 Performance Goals

- **Minimize Customer Waiting Time** – Reduce the time customers spend waiting for checkout service.
- **Increase Throughput** – Increase the number of customers served during the operating period.
- **Reduce Queue Length** – Prevent excessive queues and reduce customer delays.
- **Optimize Cashier Utilization** – Use available cashier resources efficiently without unnecessary overuse or under use.
- **Identify Bottlenecks** – Identify conditions that cause delays, long queues, or reduced system performance.
- **Compare Different Scenarios** – Analyze baseline conditions, high customer traffic, additional cashiers, and bulk-item purchases.
- **Improve Customer Service** – Provide a faster and smoother checkout experience while using available resources effectively.

3. Modeling Approach and Assumptions

3.1 Modeling Approach

The supermarket cashier checkout system is modeled using Discrete Event Simulation (DES) with Python and SimPy. The model simulates customer arrivals, cashier queues, service times, and customer departures.

Customers arrive at random intervals and request an available cashier. If all cashiers are busy, customers wait in the queue. Once a cashier becomes available, the customer's waiting time is recorded. Service time is randomly generated using an exponential distribution based on the defined mean service time.

The main performance measures are average waiting time, throughput, average queue length, and cashier utilization.

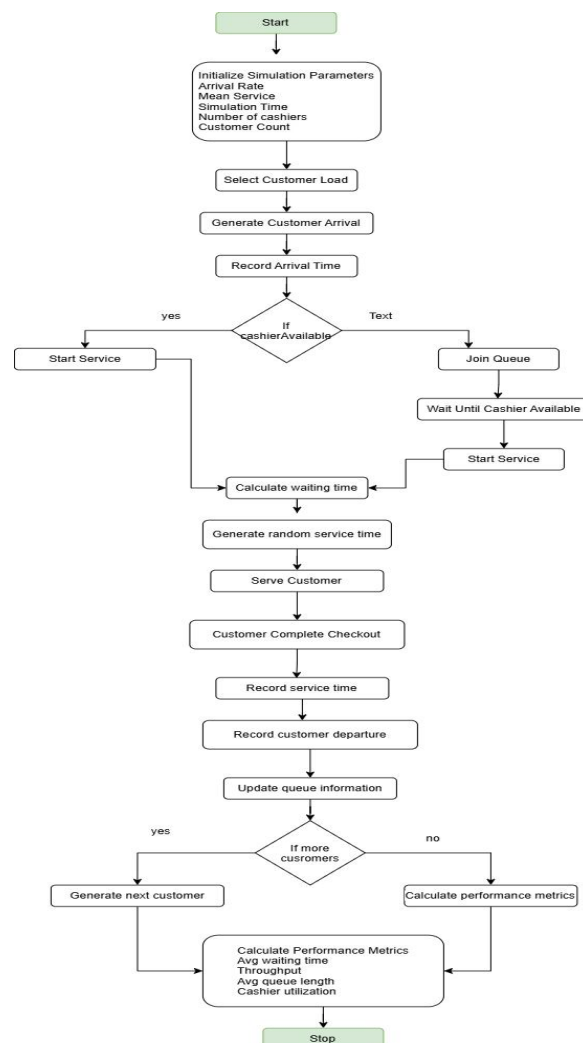


Figure 01 - Supermarket Cashier Checkout Simulation Mode

3.3 Assumptions

- Customer arrival times are random.
- Service times are random, with an average of 4 minutes.
- Each cashier serves one customer at a time.
- Customers wait when all cashiers are busy.
- The simulation runs for 480 minutes (8 hours).
- Different customer loads are tested.
- Cashier breaks, payment failures, and customer leaving are not considered.

4.Data Description and Methodology

Python and SimPy were used to generate simulation data for evaluating the performance of the supermarket checkout system under different customer loads.

The simulation uses the following input parameters:

- **Number of Cashiers** – Number of cashiers available for checkout.
- **Number of Customers** – Customer loads of 10, 18, 27, 35, 48, 56, and 66 are tested.
- **Arrival Rate** – Determines the time between customer arrivals.
- **Mean Service Time** – Average checkout service time is set to 4 minutes.
- **Simulation Time** – The simulation runs for 480 minutes (8 hours).

During the simulation, customer waiting times, service times, and queue lengths are recorded. The following performance measures are calculated:

- **Customers** – Number of customers processed.
- **Cashiers** – Number of cashiers available.
- **Average Wait** – Average time customers wait for a cashier.
- **Throughput** – Number of customers served per minute.
- **Average Queue** – Average number of customers waiting in the queue.
- **Utilization** – Proportion of cashier capacity being used.

Sample Data Set

Customers	Cashiers	AverageWait	Throughput	Average Queue	Utilization
10	5	0.91	0.0208	0.4	0.022
18	5	0.97	0.0375	1.17	0.028
27	5	1.98	0.0563	2.59	0.037
35	5	1.77	0.0729	1.06	0.06
48	5	13.44	0.1	10.48	0.103
56	5	5.99	0.1167	5.45	0.106
66	5	5.05	0.1375	6.35	0.101

Simulate data set .csv file - https://github.com/ChanuHasintha/Performance-Modelling-MP/blob/main/supermarket_results.csv

5.Detailed Analysis and Findings

The simulation results are evaluated using average waiting time, throughput, average queue length, and cashier utilization. Different customer loads are compared to understand how increasing demand affects the supermarket checkout system.

- **Average Waiting Time-** As the number of customers increases while cashier capacity remains unchanged, customers are likely to wait longer for service.
- **Throughput** - Throughput shows the number of customers served per minute. It indicates the system's ability to handle customer demand.
- **Average Queue Length-** The average queue length tends to increase when more customers arrive and the available cashier capacity remains the same.
- **Cashier Utilization-** Cashier utilization increases as customer demand increases, indicating that cashiers are occupied for a greater proportion of the simulation period.

The results help identify how increasing customer demand affects waiting time, queue length, throughput and cashier utilization. This information can be used to determine an appropriate number of cashiers for efficient supermarket operation.

6.Visualizations

Graphical representations are used to clearly present and interpret the simulation results of the supermarket checkout system.

- ❖ This all graphs are represent the performance with 8 cashiers.

Scenario

Enter number of cashiers : 8

- ✓ **Average Waiting Time for Different Customer Loads**

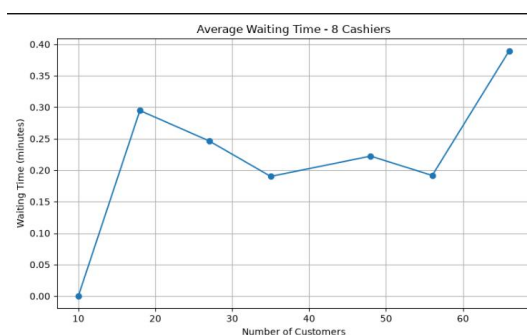


Figure 02 - Average waiting time

This graph shows how the average wait time for consumers changes as the number of customers increase.

✓ **Average Queue Length for Different Customer Loads**

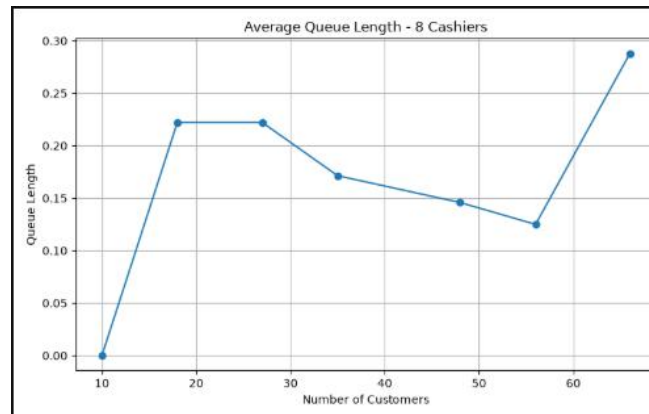


Figure 03 - Average queue length

This graph shows how the average number of customers waiting in the queue changes with customer demand.

✓ **Cashier Utilization for Different Customer Loads**

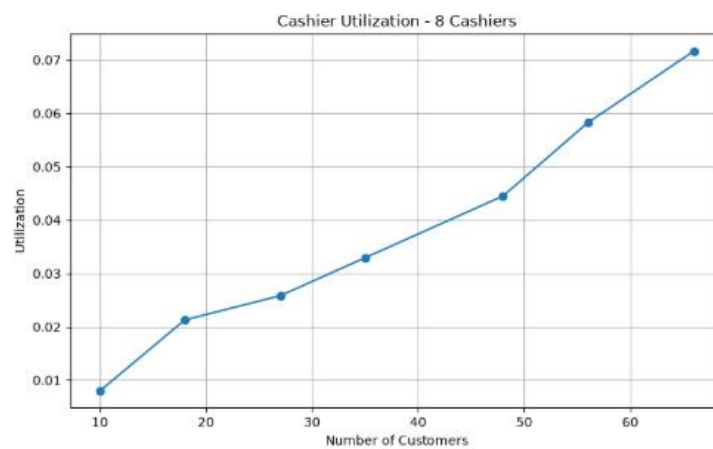


Figure 04 - Cashier Utilization

This graph shows how the utilization of cashier resources changes as the number of customers increases.

✓ Customer Throughput for Different Customer Loads

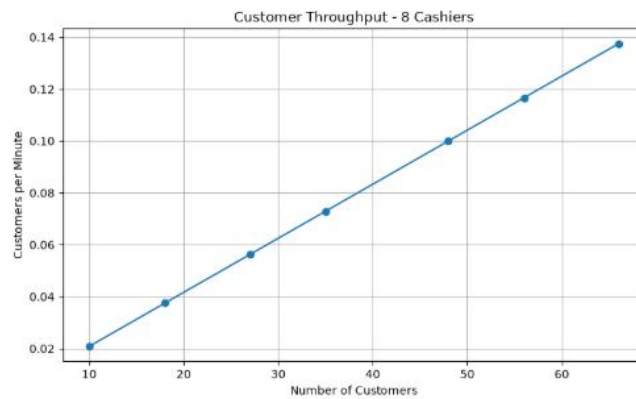


Figure 05- Customer Throughput

This graph shows the number of customers served per minute under different customer loads.

7. Limitations and Future Extensions

7.1 Limitations

- The model uses simulated data instead of real supermarket data.
- Customer arrival and service times are randomly generated.
- The simulation uses a fixed number of cashiers for each test.
- Cashier breaks, payment problems and system failures are not included.
- Customer behaviors such as changing queues or leaving before service are not modeled.

7.2 Future Extensions

- **Extend the model to include weekend and peak-hour patterns**

The model can be extended to represent different customer arrival rates during weekends and busy peak hours. This would help analyze changes in waiting time, queue length, throughput and cashier utilization under high demand conditions.

- **Use real supermarket data**

Actual data on customer arrivals, number of items and checkout service times can be collected from a supermarket. Using real world data would make the simulation more realistic and improve the accuracy of the performance analysis.

8. References

- [1] J. V. Pereira Junior, A. M. da Silva, and D. G. de Moraes, "Discrete simulation applied to queue management in a supermarket," *Independent Journal of Management & Production*, vol. 11, no. 5, pp. 1667–1684, 2020, doi: 10.14807/ijmp.v11i5.1296.
- [2] SimPy Developers, "SimPy Documentation," 2026. [Online]. Available:
- [3] Matplotlib Development Team, "Matplotlib Documentation," 2026. [Online]. Available:
- [4] C. Hasintha, "Supermarket Cashier Checkout Simulation: Simulation Code and Generated Dataset," *GitHub repository*, 2026. [Online]. Available:[https://github.com/ChanuHasintha/Performance-Modelling-MP/blob/main/supermarket_checkout_analysis_dataset.xlsx].

9. Appendix

GitHub Repositories Link - <https://github.com/ChanuHasintha/Performance-Modelling-MP.git>

```
422510839_Mp.py > ...
1  #422510839
2  #s22010004
3  #L.A.M.C.H.Karunasena
4
5  import simpy
6  import random
7  import statistics
8  import csv
9  import os
10 import matplotlib.pyplot as plt
11
12
13
14 # Simulation Parameters
15 RANDOM_SEED = 42
16 ARRIVAL_RATE = 2
17 MEAN_SERVICE = 4
18 SIM_TIME = 8 * 60
19
20 CUSTOMER_COUNTS = [10, 18, 27, 35, 48, 56, 66]
21
22 OUTPUT_FILE = "supermarket_results.csv"
23
24
25
26 # Customer Process
27
28 def customer(env, cashiers, waits, services, queues):
29
30     # Record customer arrival time
31     arrival = env.now
32
33     # Record queue length at arrival
34     queues.append(len(cashiers.queue))
35
36     # Request a cashier
37     with cashiers.request() as request:
```



```

28     def customer(env, cashiers, waits, services, queues):
29
30         # Wait until a cashier becomes available
31         yield request
32
33         # Calculate waiting time
34         wait_time = env.now - arrival
35         waits.append(wait_time)
36
37         # Generate service time using exponential distribution
38         service = random.expovariate(1 / MEAN_SERVICE)
39         services.append(service)
40
41         # Customer receives checkout service
42         yield env.timeout(service)
43
44 # Customer Generator
45
46 def generate_customers(
47     env,
48     cashiers,
49     waits,
50     services,
51     queues,
52     total
53 ):
54
55     for i in range(total):
56
57         # Generate random time between customer arrivals
58         arrival = random.expovariate(ARRIVAL_RATE)
59
60         yield env.timeout(arrival)
61
62         # Create customer process
63         env.process(
64             # Create customer process
65             customer(
66                 env,
67                 cashiers,
68                 waits,
69                 services,
70                 queues
71             )
72         )
73
74 # Run One Simulation
75 def simulate(customers, cashier_count):
76
77     # Reset random generator using fixed seed
78     random.seed(RANDOM_SEED)
79
80     # Create SimPy environment
81     env = simpy.Environment()
82
83     # Create cashier resource
84     cashiers = simpy.Resource(
85         env,
86         capacity=cashier_count
87     )
88
89     # Lists for performance measurements
90     waits = []
91     services = []
92     queues = []

```

```

101 services = []
102 queues = []
103
104 # Start customer generator
105 env.process(
106     generate_customers(
107         env,
108         cashiers,
109         waits,
110         services,
111         queues,
112         customers
113     )
114 )
115
116 # Run simulation for 480 minutes
117 env.run(until=SIM_TIME)
118
119 # Calculate average waiting time
120 avg_wait = (
121     statistics.mean(waits)
122     if waits else 0
123 )
124
125 # Calculate average queue length
126 avg_queue = (
127     statistics.mean(queues)
128     if queues else 0
129 )
130
131 # Calculate throughput
132 throughput = len(services) / SIM_TIME
133
134 # Calculate cashier utilization
135 utilization = (
136     sum(services) /
136     sum(services) /
137     (cashier_count * SIM_TIME)
138 )
139
140 return [
141     len(services),
142     avg_wait,
143     throughput,
144     avg_queue,
145     utilization
146 ]
147
148
149 # Run All Simulations
150 def run_simulations(cashier_count):
151
152     results = []
153
154     print("\n" + "=" * 75)
155     print(f"Results for {cashier_count} Cashiers")
156     print("=" * 75)
157
158     print(
159         f"{'Customers':<12}"
160         f"{'Wait':<12}"
161         f"{'Throughput':<15}"
162         f"{'Queue':<12}"
163         f"{'Utilization':<12}"
164     )
165
166     print("-" * 75)
167
168     # Run simulation for each customer load
169     for customers in CUSTOMER_COUNTS:
170

```

```

171     result = simulate(
172         customers,
173         cashier_count
174     )
175
176     results.append(result)
177
178     print(
179         f"{result[0]:<12}"
180         f"{result[1]:<12.2f}"
181         f"{result[2]:<15.4f}"
182         f"{result[3]:<12.2f}"
183         f"{result[4]:<12.3f}"
184     )
185
186     # Save results to CSV
187     save_csv(
188         results,
189         cashier_count
190     )
191
192     # Create graphs
193     create_graphs(
194         results,
195         cashier_count
196     )
197
198
199 # Save Results to CSV
200 def save_csv(results, cashier_count):
201
202     exists = os.path.exists(OUTPUT_FILE)
203
204     with open(
205         OUTPUT_FILE,
206         "a",
207         newline=""
208     ) as file:
209
210         writer = csv.writer(file)
211
212
213         if not exists:
214
215             writer.writerow([
216                 "Customers",
217                 "Cashiers",
218                 "Average Wait",
219                 "Throughput",
220                 "Average Queue",
221                 "Utilization"
222             ])
223
224         # simulation results
225         for r in results:
226
227             writer.writerow([
228                 r[0],
229                 cashier_count,
230                 round(r[1], 2),
231                 round(r[2], 4),
232                 round(r[3], 2),
233                 round(r[4], 3)
234             ])

```

```

236     print(
237         f"\nResults saved to {OUTPUT_FILE}"
238     )
239
240
241 # Create Graphs
242 def create_graphs(
243     results,
244     cashier_count
245 ):
246
247     customers = [
248         r[0] for r in results
249     ]
250
251     wait = [
252         r[1] for r in results
253     ]
254
255     queue = [
256         r[3] for r in results
257     ]
258
259     utilization = [
260         r[4] for r in results
261     ]
262
263     throughput = [
264         r[2] for r in results
265     ]
266
267     # Graph information
268     graphs = [
269
270         (
271             wait,
272             "Average Waiting Time",
273             "Waiting Time (minutes)",
274             "average_waiting_time.png"
275         ),
276
277         (
278             queue,
279             "Average Queue Length",
280             "Queue Length",
281             "average_queue_length.png"
282         ),
283
284         (
285             utilization,
286             "Cashier Utilization",
287             "Utilization",
288             "cashier_utilization.png"
289         ),
290
291         (
292             throughput,
293             "Customer Throughput",
294             "Customers per Minute",
295             "throughput.png"
296         )
297     ]
298
299     # Create each graph
300     for values, title, ylabel, filename in graphs:
301
302         plt.figure(figsize=(8, 5))

```

```

302     plt.figure(figsize=(8, 5))
303
304     plt.plot(
305         customers,
306         values,
307         marker="o"
308     )
309
310     plt.title(
311         f"{title} - {cashier_count} Cashiers"
312     )
313
314     plt.xlabel(
315         "Number of Customers"
316     )
317
318     plt.ylabel(ylabel)
319
320     plt.grid(True)
321
322     plt.tight_layout()
323
324     plt.savefig(filename)
325
326     plt.show()
327
328     print(
329         "\n4 diagrams created successfully!"
330     )

```

```

333 #Main Program
334 if __name__ == "__main__":
335
336     try:
337
338         cashier_count = int(
339             input(
340                 "Enter number of cashiers: "
341             )
342         )
343
344         # Check cashier count
345         if cashier_count <= 0:
346             raise ValueError
347
348         # Run simulations
349         run_simulations(
350             cashier_count
351         )
352
353     except ValueError:
354
355         print(
356             "Invalid input. "
357             "Please enter a positive number."
358         )
359

```

Output

Enter number of cashiers: 10

=====

Results for 10 Cashiers

=====

Customers	Wait	Throughput	Queue	Utilization
10	0.00	0.0208	0.00	0.006
18	0.00	0.0375	0.00	0.015
27	0.31	0.0563	0.56	0.023
35	0.51	0.0729	1.09	0.030
48	0.46	0.1000	0.94	0.043
56	0.39	0.1167	0.82	0.050
66	0.34	0.1375	0.70	0.058

Results saved to supermarket_results.csv